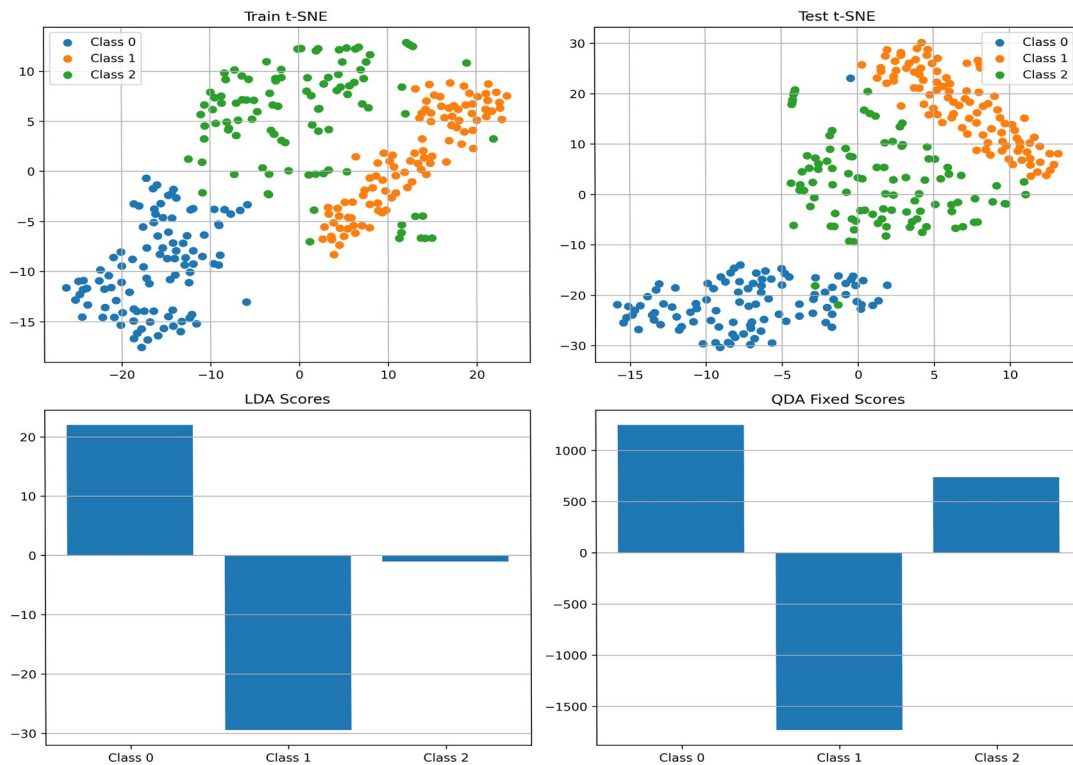


2024450_A1



- In the first two graphs, we visualized both the training and test datasets using t-SNE. This technique reduces our 728-dimensional, column-stacked image data into a 2-dimensional representation. t-SNE works by placing points close together in the lower-dimensional space if they were close in the original high-dimensional space. Because of this, the axes in the plot do not have any specific meaning — only the relative distances between points matter.
- Two bar graphs are nothing just the values of discriminant of a sample input. The higher the value of the discriminant more the probability that input should belong to that class.

About The Code

- The dataset is downloaded using **kagglehub**, which provides the raw MNIST binary files.
The custom `MnistDataLoader` class reads these binary files and converts them into:
`x_train, y_train` → training images and labels
`x_test, y_test` → test images and labels

- We keep only images of digits **0, 1, and 2** to make this a **3-class classification problem**. For each class (0, 1, 2), we randomly select **100 training images** and **100 test images**. This keeps the dataset balanced and small enough for covariance calculations.

- Each image is:

Normalized to the range [0,1]

Flattened from 28×28 into a 784-dimensional vector

Transposed so each column is one image

- For each class, we compute the **mean image vector**, representing the average digit appearance.
- **QDA** → Each class gets its own **covariance matrix**, which captures how pixel intensities vary together. Since covariance matrices can be unstable in high dimensions, we add:

$$g_k(x) = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) - \frac{1}{2} \log |\Sigma_k|$$

- **LDA** → **all classes share one covariance matrix which is mean of all of them:**

$$g_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k$$